Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2002-I-Q02] | Part A (Q1-Q5)

Answer: 154

1. For radius r, the width is $14r$. Nearest centers in adjacent rows form equilateral triangles of side $2r$, so the vertical row gap is $\sqrt{3}r$.

2. The height is $2r + 2\sqrt{3}r$, giving ratio $7/(1+\sqrt{3}) = (\sqrt{147}-7)/2$. Hence $p+q = 154$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2005-I-Q01] | Part A (Q1-Q5)

Answer: 942

1. If the small radius is r, the centers form a regular hexagon of side and circumradius $2r$. Internal tangency gives $2r + r = 30$, so $r = 10$.

2. The area is $K = \pi(900 - 6 \cdot 100) = 300\pi$, between 942 and 943. Its floor is 942.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2009-II-Q05] | Part A (Q1-Q5)

Answer: 032

1. Place the first small-circle center at $(0, 7)$ and the equal pair at $(\pm 4\sqrt{3}, -4)$. Equal distances to that pair force the fourth center to be $(0, y)$.

2. If its radius is r, tangency to the top circle requires $y = 4 - r$ or $10 + r$. The latter is impossible in the equation for either lower circle, so use $y = 4 - r$. Then $48 + (8 - r)^2 = (r + 2)^2$.

3. Solve $r = 27/5$, giving 32.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2000-II-Q10] | Part B (Q6-Q10)

Answer: 647

1. For inradius r, the four half-angle tangents are $r/19, r/26, r/37, r/23$, and the half-angles sum to 180°.

2. Pairing opposite half-angles gives $56r/(703 - r^2) = 49r/(r^2 - 598)$. Thus $105r^2 = 56 \cdot 598 + 49 \cdot 703 = 67935$.

3. So $r^2 = 647$. The corresponding positive half-angles give the required 180° sum; uniqueness follows since that sum increases strictly with r.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2005-II-Q08] | Part B (Q6-Q10)

Answer: 405

1. The small-center separation is 14, so $(R-4)+(R-10)=14$ and the outer radius is $R=14$. Its center lies ten units from the radius-four center and four from the other.

2. Signed distance to the common tangent varies affinely along the center line. Thus the outer-center distance is $4+(10/14)(10-4)=58/7$. Chord length is $2\sqrt{14^2-(58/7)^2}=8\sqrt{390}/7$.

3. Since 390 is squarefree and $8,7$ coprime, the answer is 405.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2006-II-Q09] | Part B (Q6-Q10)

Answer: 027

1. Internal similarity centers divide center distances in radius ratios, giving tangent-line intercepts $(4,0)$ and $(16,0)$. Right triangles yield slopes $1/\sqrt{15}$ and $-1/\sqrt{3}$.

2. Equate $(x-4)/\sqrt{15} = -(x-16)/\sqrt{3}$, so $x = (4+16\sqrt{5})/(1+\sqrt{5}) = 19-3\sqrt{5}$. The requested sum is 27.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2007-I-Q09] | Part B (Q6-Q10)

Answer: 737

1. The hypotenuse is 34. Circle centers share the same perpendicular distance r from it, so their tangency-foot separation is $2r$. At the two ends, tangent distances are $r\tan(A/2)$ and $r\tan(B/2)$.

2. Half-angle formulas give $\tan(A/2) = 16/(34+30) = 1/4$ and $\tan(B/2) = 30/(34+16) = 3/5$. Hence $34 = r(1/4 + 2 + 3/5) = 57r/20$.

3. Thus $r = 680/57$ and $p + q = 737$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2014-II-Q08] | Part B (Q6-Q10)

Answer: 254

1. Let the smaller radius be s, so D has radius $3s$. Set the outer center at $(0,0)$, $A=(-2,0)$, the center of D at $(-2+3s,0)$, and the center of E at (x,s), choosing the upper orientation.

2. Internal tangency gives $x^2=4-4s$. External tangency to D gives $(x+2-3s)^2+s^2=16s^2$. Eliminate x^2 to get $x(2-3s)=3s^2+8s-4$.

3. Squaring this equation and using $x^2=4-4s$ yields $s^2(9s^2+84s-44)=0$. The only positive root is $s=(-14+4\sqrt{15})/3$. It has $0<3s<2$, and the positive value of x satisfies the unsquared equation, so it is valid.

4. Thus the requested radius is $3s=4\sqrt{15}-14=\sqrt{240}-14$, giving 254.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2022-I-Q08] | Part B (Q6-Q10)

Answer: 378

1. For the vertex-A circle, let its center distance from A be d>0. Its radius is d/2 by the 30-degree half-angle. Internal tangency is |18-d|=18-d/2. If d<=18 this forces d=0, impossible; hence d>18 and d-18=18-d/2, giving d=24 and radius12.

2. Put the circumcenter at $(0,0)$ and $A=(0,18)$. The small-circle centers are $(0,-6)$, $(3\sqrt{3},3),(-3\sqrt{3},3)$. The last two circles meet on $x=0$ at $y=3\pm\sqrt{117}$.

3. The smaller triangle has circumradius $\sqrt{117}-3$. Its side equals $\sqrt{3}(\sqrt{117}-3)=\sqrt{351}-\sqrt{27}$, giving $351+27=378$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2022-II-Q07] | Part B (Q6-Q10)

Answer: 192

1. The two external tangent lines meet at the external homothety center P. If P is beyond the smaller circle, its distances to the large and small centers have ratio $4:1$ and difference 30, so they are $40, 10$.

2. The half-angle t at P satisfies $\sin t = 24/40 = 3/5$, hence $\tan t = 3/4$. The common tangent at the circles touch point is perpendicular to the center line; its distance from P is $40 - 24 = 16$.

3. The triangle has height 16 and half-base $16 \cdot 3/4 = 12$. Its area is $16 \cdot 12 = 192$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2024-I-Q08] | Part B (Q6-Q10)

Answer: 197

1. Set $k = \cot(B/2) + \cot(C/2)$. A chain of N circles of radius ρ occupies base length $BC = \rho k + 2\rho(N-1)$, because the end-center offsets are $\rho\cot(B/2)$ and $\rho\cot(C/2)$.

2. Thus $34k + 476 = k + 4046$, whence $k = 1190/11$ and $BC = 45696/11$.

3. The incircle is the case $N = 1$, so $r = BC/k = 192/5$. Therefore $m + n = 197$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2025-II-Q06] | Part B (Q6-Q10)

Answer: 293

1. Place the large center at $(0,0)$ and take $B=(15,0), A=(9,0), C=(-15,0), D=(9,12)$. Rectangle vertices have coordinates $(9\pm a, \pm b)$ with $a^2+b^2=36$, and the specified sides make GF top and GH left.

2. The equal triangle areas are $a(12-b)=b(24-a)$, so $a=2b$ and $5b^2=36$.

3. Rectangle area $4ab=8b^2=288/5$, hence $m+n=293$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2002-II-Q14] | Part C (Q11-Q15)

Answer: 098

1. The tangency to AM occurs at A, so $AO = 19$. If T is the contact point on PM, equal tangents from M give $MT = MA$.

2. Let $a = MA$. Right triangles POT, PMA are similar with ratio $19/a$. The smaller perimeter is $PO + OT + PT = PA + PT = 152 - 2a$. Therefore $19/a = (152 - 2a)/152$, which gives $(a - 38)^2 = 0$.

3. The ratio is $1/2$, so $PO = PM/2$ and $PA = PO + 19$. Using $PA + PM = 114$ gives $PM = 190/3$, $PO = 95/3$. Thus the answer is 98.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2002-II-Q15] | Part C (Q11-Q15)

Answer: 282

1. Two circles meeting twice must lie on the same side of each common tangent; the common point above the x-axis puts both centers above it. Their centers lie on one angular bisector. Write them as (x_i, tx_i) with radius $tx_i > 0$.

2. The common point gives $x_i^2 - (18 + 12t)x_i + 117 = 0$. Thus $x_1 x_2 = 117$ and $r_1 r_2 = 117t^2 = 68$. An obtuse upper sector would have $t < -1$, incompatible with $t^2 = 68/117 < 1$, so choose $t > 0$.

3. The line slope is $m = 2t/(1 - t^2) = 12\sqrt{221}/49$. Hence $a + b + c = 12 + 221 + 49 = 282$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2004-II-Q12] | Part C (Q11-Q15)

Answer: 134

1. The trapezoid height is $\sqrt{25-1}=\sqrt{24}$. Equal tangency distances to each equal-radius pair put the inner center on the symmetry axis. If its radius is r, its distances from the two base lines are $\sqrt{r^2+6r}$ and $\sqrt{r^2+4r}$.

2. Their sum is $\sqrt{24}$. Isolating and squaring gives $\sqrt{24(r^2+4r)}=12-r$, then $23r^2+120r-144=0$.

3. The positive root is $(-60+48\sqrt{3})/23$; it is positive, below 12, and satisfies the original radical equation. The parameter sum is 134.

4. The root gives a circle wholly inside the trapezoid: both base distances exceed r, and its distance to either leg is $(3\sqrt{24}-\sqrt{r^2+6r})/5>r$. Internal tangency to one member of a base pair is impossible: the two equal corner circles are externally tangent to each other, so such a circle touching the other member would have to touch the base line, contradicting containment with positive radius. Thus all four distance sums used above are valid.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2007-II-Q15] | Part C (Q11-Q15)

Answer: 389

1. Let original inradius and circumradius be r, R, and the common small radius x. The three corner-circle centers form the triangle homothetic to the original about its incenter, with ratio $(r-x)/r$. Their circumradius is $R(r-x)/r$.

2. The fourth center is that circumcenter, distance $2x$ from all three. Heron gives area 84, so $r=4, R=65/8$. Hence $2x=(65/8)(4-x)/4$.

3. Solve $129x=260$, giving $x=260/129$ and answer 389.

4. The original triangle is acute because $15^2<13^2+14^2$. Its circumcenter, and therefore the fourth center, is inside the homothetic corner-center triangle. Every point of that triangle is at least x from each original side, so the fourth radius-x disk is inside the original triangle; the construction is attainable.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2008-II-Q11] | Part C (Q11-Q15)

Answer: 254

1. Use $B=(0,0), C=(56,0), A=(28,96)$. Half-angle tangent at either base vertex is $(24/25)/(1+7/25)=3/4$. Thus the two centers are $(104/3,16)$ and $(4q/3,q)$, where q is the unknown radius.

2. External tangency gives $((104-4q)/3)^2+(16-q)^2=(16+q)^2$, or $(26-q)^2=36q$. Hence $q=44\pm 6\sqrt{35}$.

3. The large root cannot fit inside the triangle (its diameter exceeds the altitude), so $q=44-6\sqrt{35}$ and $m+nk=44+210=254$.

4. The triangle has area 2688 and semiperimeter 128, hence inradius 21. The smaller root satisfies $0<44-6\sqrt{35}<21$. A base-corner circle of radius below the inradius is contained inside the opposite side too, so this root gives a valid circle.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2015-II-Q15] | Part C (Q11-Q15)

Answer: 129

1. The homothety centered at A with ratio -4 maps D to E. Thus $AE = 4AD$, and equal triangle areas require distances $d(B, \ell) = 4d(C, \ell)$. The same-side condition fixes the signed ratio as positive.

2. Place centers at $(0, 1), (4, 4)$, so $B = (0, 0), C = (4, 0), A = (4/5, 8/5)$. Write the line as $u(x - 4/5) + v(y - 8/5) = 0$. The distance ratio gives $-(4u + 8v) = 4(16u - 8v)$, hence $v/u = 17/6$ and $\ell : 6x + 17y = 32$.

3. The distances from B and the small-circle center to this line are $32/\sqrt{325}$ and $15/\sqrt{325}$. Therefore its small-circle chord is $AD = 2\sqrt{1 - 225/325} = 20/\sqrt{325}$.

4. The common area is $\frac{1}{2}(32/\sqrt{325})(20/\sqrt{325}) = 64/65$. Hence the requested sum is 129.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2019-I-Q11] | Part C (Q11-Q15)

Answer: 020

1. Put base $BC = a$, equal sides b, altitude $h = \sqrt{b^2 - a^2/4}$. The inradius is $r = ah/(a + 2b)$, base-exradius $r_A = ah/(2b - a)$, and other exradii equal h. Coordinates may be chosen with incenter $(0, r)$ and excenters $(0, -r_A)$, $(\pm b, h)$.

2. If the target circle radius is ρ, internal tangency forces $\rho = r + 2r_A$, and external tangency gives $b^2 + (h - r)^2 = (h + r + 2r_A)^2$. Therefore $b^2 = 4(r + r_A)(h + r_A)$. Substitute the radii and $h^2 = (4b^2 - a^2)/4$ to get $b^2 = 8ab^2/(2b - a)$, hence $2b = 9a$.

3. Integer sides are $a = 2t, b = 9t$ for $t \geq 1$, giving perimeter $20t$. At $t = 1$ all tangency formulas hold with positive radii, so the minimum is 20.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2020-II-Q13] | Part C (Q11-Q15)

Answer: 060

1. Equal tangents at each vertex give tangent lengths $3, 2, 4, 2, 4$. If the inradius is r, the half-angle at a vertex of tangent length t has tangent r/t. The five half-angles sum to 270°.

2. Put $\beta = \arctan(r/2)$ and $\gamma = \arctan(r/4)$. Then $r/3 = \cot(2\beta + 2\gamma)$. With $T = \tan(\beta + \gamma) = 6r/(8 - r^2)$, this gives $2rT = 3(1 - T^2)$ and hence $5r^4 - 84r^2 + 64 = 0$.

3. The candidates are $r^2 = 16$ and $4/5$. The latter has every half-angle below 45°, so cannot sum to 270°. For $r = 4$ the half-angles have the required sum (the sum is strictly increasing in r).

4. The semiperimeter is 15, so area $= 15r = 60$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2021-I-Q13] | Part C (Q11-Q15)

Answer: 672

1. Let the third center and radius be O, r, and the required center distance be d. External tangency makes the difference of the powers of O equal to $(r+961)^2-961^2-[(r+625)^2-625^2]=672r$.

2. The power difference is a linear function vanishing on AB, with gradient magnitude $2d$. Hence its absolute value at O is $2d\,\mathrm{dist}(O,AB)$.

3. The 120° chord gives distance $r\cos 60^\circ = r/2$. Thus $672r = dr$, giving $d = 672$.

Source / 来源